

BERNSTEIN'S PROBLEM ON MINIMAL SURFACES

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ABSTRACT. Minimal surfaces—local minimizers of area—are amongst the most classical objects of mathematics: 1-dimensional minimal surfaces are geodesics, and 2-dimensional minimal surfaces model the behavior of soap bubbles. A very natural problem is to classify minimal surfaces in the codimension 1 Euclidean case (n -dimensional minimal surfaces in $\mathbb{R}^{n+1}$). Geometric intuition tells us that hyperplanes are minimal surfaces, but in 1917 S. Bernstein posed the converse question: are n -dimensional hyperplanes the only minimal surfaces in $\mathbb{R}^{n+1}$? I will explain the answer to this question, which for deep geometric and analytic reasons (and the quadratic formula) turns out to be yes for $n \leq 7$ and no for $n \geq 8$.

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1. INTRODUCTION: BERNSTEIN'S PROBLEM

The topic of my talk will be on Bernstein's problem on minimal surfaces. To begin, I will describe imprecisely what the problem is.

Roughly speaking, minimal surfaces are surfaces which locally minimize area. A useful physical model to keep in mind is that of soap films. Way back in 1762, Lagrange discovered that for a surface given by a parametric equation $z = u(x, y)$, being minimal is equivalent to solving the *minimal surface equation* (MSE)

$$0 = \operatorname{div} \left(\frac{u_x}{\sqrt{1 + u_x^2 + u_y^2}}, \frac{u_y}{\sqrt{1 + u_x^2 + u_y^2}} \right) = \operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right).$$

Obviously, if ∇u is constant, then its graph $\Gamma_u = \{z = u(x, y)\}$ is a minimal surface. In this case, u is affine (i.e., $u(x, y) = ax + by + c$), so that its graph is a plane. Moreover, this particular function u is defined on all of $\mathbb{R}^2$. This prompts a very natural question, which came to be known as Bernstein's problem:

Question 1.1. *Let $u: \mathbb{R}^n \rightarrow \mathbb{R}$ be an entire function satisfying the MSE. Then, must u be affine?*

Compare this to Liouville's theorem:

Theorem 1.2. *Let $u: \mathbb{R}^n \rightarrow \mathbb{R}$ be an entire function satisfying $\Delta u = 0$. If u is bounded, then u must be constant.*

Liouville's theorem holds for all n , but it turns out that the answer to Bernstein's problem is yes for $n \leq 7$, and no for $n \geq 8$. For $n = 8$ there exist counterexamples: functions $u: \mathbb{R}^8 \rightarrow \mathbb{R}$ which satisfy the MSE but are not flat.

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The $n = 2$ case was solved by Bernstein (1917), and remained open until the 1960s, when in a flurry of activity the $n \leq 7$ cases were resolved by work of Fleming (1962, $n = 2$ by a new strategy), de Giorgi (1965, $n = 3$ by refining Fleming's strategy), Almgren (1966, $n = 4$), and Simons (1968, $n = 5, 6, 7$), and then the $n \geq 8$ counterexample was found by Bombieri, de Giorgi, and Giusti (1969).

The goal of my talk will be to explain this wonderful fact, which touches on a wide variety of tools in analysis, geometry, and PDE.¹

2. THE AREA FUNCTIONAL

Recall that if $\Omega \subseteq \mathbb{R}^n$ is a bounded domain, and $u \in C^2(\overline{\Omega})$, we can define its *graph*

$$\Gamma := \Gamma_u = \{(x, u(x)) \in \Omega \times \mathbb{R} : x \in \Omega\}.$$

If u is a C^k function, then Γ_u is a C^k submanifold of $\mathbb{R}^{n+1}$. Assuming u is at least C^1 , the *area* of this graph is given by

$$\mathcal{A}(u) = \int_{\Gamma} 1 = \int_{\Omega} \sqrt{1 + |\nabla u|^2}.$$

This formula generalizes the usual arc length formula in the case $n = 1$:

$$\text{len}(y(x)) = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

This formula has the following interpretation: the (upwards pointing) unit normal Γ_u is

$$\nu = \frac{(-\nabla u, 1)}{\sqrt{1 + |\nabla u|^2}}.$$

If we normalize this vector to have its last coefficient equal to 1, the L^2 -norm of the resulting vector in the first n components is the “stretch factor” of the area.

As with above, we have first and second variation formulae for the area:

Proposition 2.1 (First variation formula). *Let $u \in C^2(\overline{\Omega})$ and $\varphi \in C_c^\infty(\Omega)$. Then, the first variation of $\mathcal{A}$ at u in the direction of φ is*

$$\delta_\varphi \mathcal{A}(u) = - \int_{\Omega} \varphi \operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = \langle \varphi, -H \rangle,$$

where H is the mean curvature

$$H = -\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right).$$

¹The reader interested in details is encouraged to read the author's senior thesis, which is an expository article on Bernstein's problem.

We say that u is a *stationary point for the area*, or Γ_u is a *minimal surface*, if $H = 0$. Just as before, the first variation formula tells us that the mean curvature is the gradient of the area (and hence, mean curvature flow is the gradient flow of the area functional).

To write the second variation formula, we finally have to start doing some geometry. Note that we can think about $\Gamma = \Gamma_u$ in two ways: as a graph over Ω , or as a submanifold in its own right. In the former case, note that we have been taking variations in the e_{n+1} direction. But, the more geometrically natural way to take variations is along the unit normal ν . Indeed, if we had done this all along, we would have considered variations of the form $\Gamma + t\varphi\nu$ (this is well-defined for small t , as long as Γ is embedded), and gotten the first variation formula

$$\delta_\varphi \mathcal{A}(\Gamma_u) = - \int_\Gamma \varphi H_\Gamma,$$

where $H_\Gamma = \operatorname{div}_\Gamma \nu$, where $\operatorname{div}_\Gamma$ is the divergence along Γ and $\varphi \in C_c^\infty(\Gamma)$. Here, $H_\Gamma = \operatorname{div}_\Gamma \nu$. In this formulation, we can write the second variation formula fairly easily.

Proposition 2.2 (Second variation formula). *Let $\Gamma^n \subseteq \mathbb{R}^{n+1}$ be an embedded hypersurface with unit normal ν , and $\varphi \in C_c^\infty(\Gamma)$. The second variation of area in the direction of φ is*

$$\delta_\varphi^2 \mathcal{A}(\Gamma) = \int_\Gamma |\nabla_\Gamma \varphi|^2 - |A_\Gamma|^2 \varphi^2 = - \langle \varphi, (\Delta_\Gamma + |A_\Gamma|^2) \varphi \rangle,$$

where $A_\Gamma = \nabla_\Gamma \nu$ is the second fundamental form.

This means that stability for minimal hypersurfaces is equivalent to either the following quantitative criterion:

$$0 \leq \lambda_1(\Delta_\Gamma + |A_\Gamma|^2) = \inf_{\varphi \in C_c^\infty(\Gamma) \setminus \{0\}} \frac{\int_\Gamma |\nabla_\Gamma \varphi|^2 - |A_\Gamma|^2 \varphi^2}{\int_\Gamma |\varphi|^2}$$

or the following inequality for all $\varphi \in C_c^\infty(\Gamma)$:

$$\int_\Gamma |A_\Gamma|^2 \varphi^2 \leq \int_\Gamma |\nabla_\Gamma \varphi|^2.$$

In our case, one can actually prove that minimal graphs are stable (in fact, area-minimizing), so we can utilize this inequality.

A note on the second fundamental form, which I will mostly black-box for the sake of this talk. The second fundamental form is an $n \times n$ matrix at each point, which intuitively is a measure of how curved a submanifold is relative to the ambient space (which in our case is $\mathbb{R}^{n+1}$). For our purposes, it will suffice to note that the second fundamental form of Γ_u is (non-uniformly) comparable to the Hessian of u :

$$\frac{|D^2 u|}{(1 + |\nabla u|^2)^{1/2}} \leq |A_\Gamma| \leq \frac{|D^2 u|}{(1 + |\nabla u|^2)^{3/2}}.$$

Hence, for our purposes it will suffice to note that

$$|A_\Gamma| \equiv 0 \implies D^2 u \equiv 0 \implies u \text{ is affine} \iff \Gamma_u \text{ is a flat hyperplane.}$$

To illustrate how this works, let us prove Bernstein's theorem for $n = 2$, black-boxing a technical fact:

Theorem 2.3 (Cutoff functions for $n = 2$). *Let $u: \mathbb{R}^2 \rightarrow \mathbb{R}$ solve the MSE, so that $\Gamma = \Gamma_u$ is a minimal surface. Then, the functions $\eta_N \in C_c^{0,1}(\Gamma)$ defined by*

$$\eta_N(x) = \begin{cases} 1 & |x| \in [0, e^N] \\ 2 - \frac{1}{N} \log |x| & |x| \in [e^N, e^{2N}] \\ 0 & |x| \in [e^{2N}, \infty) \end{cases}$$

satisfy $\eta_N \nearrow \mathbb{1}_\Gamma$ and $\|\nabla_\Gamma \eta_N\|_{L^2(\Gamma)} \searrow 0$.

Remark 2.4. This fact crucially uses the fact that minimal graphs are locally minimizing, in the following way. By *locally minimizing* we mean the following property: for all $R > 0$, $B_R \cap \Gamma$ has the least area of all hypersurfaces sharing the same boundary. This entails that minimal graphs satisfy a volume growth estimate: for $\Gamma = \Gamma_u$ with $u: \mathbb{R}^n \rightarrow \mathbb{R}$, there exists a $C > 0$ such that

$$|B_R \cap \Gamma| \leq CR^n.$$

To see this, simply note that the boundary $\Gamma \cap \partial B_R$ cuts the sphere ∂B_R into two pieces, but by the minimizing property $\Gamma \cap B_R$ has less area than both pieces. The two pieces of the sphere must have area totaling the area of $\partial B_R = \alpha_n R^n$, and hence $\Gamma \cap B_R$ has area at most $\frac{1}{2} \alpha_n R^n$. (Here $\alpha_n = \mathcal{H}^n(\partial B^{\mathbb{R}^{n+1}}(0, 1))$.)

Note that the order of volume growth need not relate to the dimension of the surface: the submanifold $\bigcup_{n \in \mathbb{Z}} \{n\} \times \mathbb{R}^2$ is a 2-manifold with cubic area growth, and the submanifold $S^1 \times \mathbb{R} \subseteq \mathbb{R}^3$ is a 2-manifold with linear area growth.

This immediately implies Bernstein's theorem for $n = 2$: taking η_N in the stability inequality gives

$$\int_\Gamma |A_\Gamma|^2 = \lim_{N \rightarrow \infty} \int_\Gamma |A_\Gamma|^2 \eta_N^2 \leq \lim_{N \rightarrow \infty} \int_\Gamma |\nabla_\Gamma \eta_N|^2 = 0,$$

and hence $A_\Gamma \equiv 0$. In fact, basically the same argument gives us the $n = 2$ case of the more general *stable Bernstein problem*:

Question 2.5. *Let $\Sigma^n \subseteq \mathbb{R}^{n+1}$ be a complete stable minimal hypersurface with unit normal ν . Then, must Σ be a hyperplane?*

Very recently, this problem has been fully resolved with the same dimension cutoff (shifted by 1) as the classical Bernstein problem:

- (1) $n \geq 7$: counterexamples from Bombieri–de Giorgi–Giusti (1969).²
- (2) $n = 2$: independently by do Carmo–Peng (1979), Fischer-Colbrie–Schoen (1980), and Pogorelov (1981);
- (3) $n = 3$: Chodosh–Li (2024) and another proof Chodosh–Li (2023);
- (4) $n = 4$: Chodosh–Li–Minter–Stryker (2026);
- (5) $n = 5$: Mazet (preprint 2024);

²The dimension shift happens because the leaves of the Hardt–Simon foliation are minimizing hypersurfaces in $\mathbb{R}^8$.

(6) $n = 6$: Han Hong, Haizhong Li, Gaoming Wang (preprint 2026).

We remark that many of the ideas in the graphical Bernstein problem apply to the stable setting, except for the volume growth bounds. Proving that such volume growth bounds hold even in the stable minimal setting has been the main technical hurdle of solving the stable Bernstein problem.

3. FLEMING'S REDUCTION TO MINIMIZING CONES

We will now discuss the geometric ideas which go in to the proof of Bernstein's theorem. We first introduce some terminology.

Definition 3.1. A set $C \subseteq \mathbb{R}^{n+1}$ is called a *cone* if $C = \bigcup_{r \geq 0} rC$: that is, for all $x \in C$ and $r \geq 0$, $rx \in C$.

Given a set $\Sigma \subseteq S^n \subseteq \mathbb{R}^{n+1}$, the *cone over* Σ is the set

$$C(\Sigma) = \bigcup_{r \geq 0} r\Sigma.$$

Given a cone $C \subseteq \mathbb{R}^{n+1}$, the *link of* C is the set

$$N(C) = C \cap S^n.$$

Clearly these two operations are inverses of one another.

We say that a cone C is *smooth* if the link $N(C)$ is a smooth submanifold of S^n . Equivalently, C is smooth if its singular set is exactly $\{0\}$.

If C is a smooth cone over a link $\Sigma^k \subseteq S^n$, then $C \setminus \{0\}$ is a smooth $(k+1)$ -dimensional submanifold of $\mathbb{R}^{n+1}$. Therefore, we say that for C a *smooth cone*, C is *minimal/stable/locally minimizing* if $C \setminus \{0\}$ is minimal/stable/locally minimizing.

Example 3.2. Let us now consider the function $u(x, y) = |x|^2 - |y|^2$ on $\mathbb{R}^2$, and let $\Gamma := \Gamma_u$ be its graph. Now suppose we rescale the graph by the map $\xi \mapsto \lambda\xi$ for $\lambda > 0$, which maps $(\xi, u(\xi)) \mapsto (\lambda\xi, \lambda u(\xi))$. Writing $\zeta = \lambda\xi$, we get that the rescaled graph

$$\lambda\Gamma = \{(\zeta, \lambda u(\lambda^{-1}\zeta)) : \zeta \in \mathbb{R}^2\},$$

and hence $\lambda\Gamma = \Gamma_{u_\lambda}$, where $u_\lambda = \lambda u(\lambda^{-1}\cdot)$.

Concretely computing, we get that

$$u_\lambda(x) = \lambda \left(|\lambda^{-1}x|^2 - |\lambda^{-1}y|^2 \right) = \lambda^{-1} (|x|^2 - |y|^2) = \lambda^{-1}u.$$

Sending $\lambda \rightarrow \infty$, we get that $u_\lambda \rightarrow 0$, and hence $\lambda\Gamma \rightarrow \mathbb{R}^2$ converges to a plane—in fact, the tangent plane of u at 0.

Sending $\lambda \rightarrow 0$, however, the situation is a little more delicate, since u_λ blows up at points where $|x_1| \neq |x_2|$. But, we can still take the limit geometrically:

$$\lambda\Gamma = \{z = \lambda^{-1}(x^2 - y^2)\} = \{\lambda z = x^2 - y^2\},$$

and hence taking $\lambda \rightarrow 0$ we get that

$$\lambda\Gamma \rightarrow \{(x, y, z) \in \mathbb{R}^3 : x^2 - y^2 = 0\} = \{(x, y) : |x| = |y|\} \times \mathbb{R},$$

which is the union of the two planes $\{x = y\}$ and $\{x = -y\}$.

Now let us return to the general case: without loss of generality, suppose we have a function $u: \mathbb{R}^n \rightarrow \mathbb{R}$ such that $u(0) = 0$, or equivalently $0 \in \Gamma := \Gamma_u$.

The *blowup cone* at 0 is the limit $\lim_{\lambda \rightarrow \infty} \lambda \Gamma$. The *blowdown cone* is the limit $\lim_{\lambda \rightarrow 0} \lambda \Gamma$.

Fleming's contribution was to provide the following proof strategy:

- (1) Construct a blowdown cone $\lambda \Gamma \rightarrow \Gamma_\infty$, and show that Γ nonflat minimal graph implies Γ_∞ is a nonflat minimizing cone.
- (2) Show that every minimizing cone $C^n \subseteq \mathbb{R}^{n+1}$ is flat.
- (3) Contradiction, therefore Γ must have been flat.

Theorem 3.3 (Fleming (1962)). *Every minimizing cone $C^2 \subseteq \mathbb{R}^3$ is flat.*

Hence, Fleming provided a new proof of Bernstein's theorem for $n = 2$.

Recall the example above; this time let us consider $u: \mathbb{R}^p \times \mathbb{R}^p \rightarrow \mathbb{R}$ with $u(x, y) = |x|^2 - |y|^2$. Then, $\Gamma = \Gamma_u$ has blowdown equal to

$$\{(x, y) \in \mathbb{R}^p \times \mathbb{R}^p: |x|^2 = |y|^2\} \times \mathbb{R}.$$

Let us denote this cone

$$C_p = \{(x, y) \in \mathbb{R}^p \times \mathbb{R}^p: |x|^2 = |y|^2\}$$

the *Simons cone*. Note that it is a $(2p - 1)$ -dimensional cone in $\mathbb{R}^{2p}$. The blowdown of Γ_u is an example of a *cylindrical cone*: a cone of the form $C = C' \times \mathbb{R}^l$ for some $l \geq 1$.

It turns out that all minimal graphs are cylindrical, with $l = 1$.

Theorem 3.4 (De Giorgi (1965)). *Let C be a blowdown of Γ . Then, $C = C' \times \mathbb{R}$, where $C' \subseteq \mathbb{R}^n$ is a minimizing hypercone.*

This augments our above strategy: Fleming's cone theorem now proves Bernstein for $n = 3$.

4. STEP A: CACCIOPPOLI SETS AND MONOTONICITY

In the toy case $u(x, y) = |x|^2 - |y|^2$, we could utilize homogeneity of u to explicitly compute the blowdown cone, but in general the business of taking limits of geometric objects is quite subtle.

Recall that in PDE theory, a standard technique for finding smooth solutions to a PDE is as follows:

- (1) (Relaxation) Find a suitable class of weak/distributional functions which solve the PDE in a weak sense.
- (2) (Existence) Use abstract functional analytic methods to prove existence of solutions in the weak setting.
- (3) (Regularity) Prove regularity theorems to show that weak solutions are in fact strong solutions.

We perform the same procedure, but this time in a geometric setting.

4.1. Relaxation. Our first task is to find a suitable class of generalized surfaces to which we could apply functional analysis.

The most general approach is to use the theory of *currents*. The idea is as follows: suppose $\Sigma^k \hookrightarrow \mathbb{R}^n$ is a smooth submanifold. Then, for a compactly supported k -form $\alpha \in \Omega_c^k(\mathbb{R}^n)$, integration defines a linear functional $[\Sigma] \in (\Omega_c^k(\mathbb{R}^n))^*$ by

$$[\Sigma] : \alpha \mapsto \int_{\Sigma} \alpha.$$

For $k = 0$, 0-manifolds are discrete sets of points and 0-forms are smooth functions, so we recover the Dirac delta, understood as a distribution in the usual sense. Now, since $\Omega_c^k(\mathbb{R}^n)$ is a Frechét space (in particular, a union of countably many Banach spaces), so we can adapt Banach–Alaoglu to prove compactness theorems for sequences of currents, provided that we have uniform bounds on compact sets. But, this is very hard: the main struggle is proving that the limits are represented by geometric objects of the right type and dimension.

In our setting, we only need oriented codimension 1 currents, or *Caccioppoli sets*, and in this case the analysis is much simpler.

Suppose $\Gamma^n \subseteq \mathbb{R}^{n+1}$ is an oriented hypersurface with unit normal ν , such that we can define $E \subseteq \mathbb{R}^{n+1}$ open with $\partial E = \Gamma$, such that ν is the inward pointing normal. The primary case we will have in mind is $\Gamma = \Gamma_u$ and $E = \text{epi } u = \{(x, z) \in \mathbb{R}^{n+1} : z > u(x)\}$ the epigraph.

Definition 4.1 (Variation measure). Let $u \in L_{\text{loc}}^1(\mathbb{R}^{n+1})$. Its *variation* is the measure

$$|Du|(W) = \sup \left\{ \int_W u \operatorname{div} \psi : \psi \in C^0(W, \mathbb{R}^{n+1}), \|\psi\|_{L^\infty} \leq 1 \right\}.$$

A function u is of *bounded variation* ($u \in \text{BV}(\mathbb{R}^{n+1})$) if $|Du|$ is finite, and of *locally bounded variation* ($u \in \text{BV}_{\text{loc}}(\mathbb{R}^{n+1})$) if $|Du|$ is finite on compact sets.

The connection with geometry comes from the following observation. Suppose $u = \mathbb{1}_E$. Then, if $W \subseteq \mathbb{R}^{n+1}$ is compact with smooth boundary, then for all $\psi \in C_c^0(\mathbb{R}^{n+1}, \mathbb{R}^{n+1})$,

$$\int_W \mathbb{1}_E \operatorname{div} \psi = \int_E \operatorname{div} \psi = \int_{\partial E} \langle \psi, \nu \rangle \leq \|\psi\|_{L^\infty} \operatorname{vol}^n(\partial E \cap W).$$

Hence, it follows that $|D\mathbb{1}_E|(W) \leq \operatorname{vol}^n(\partial E \cap W)$; it turns out that this is an equality (take $\psi = \eta\nu$ for η a compactly supported test function).

Therefore, we can see that the *perimeter of E in W* is precisely the variation of $\mathbb{1}_E$ on W :

$$P(E, W) := |D\mathbb{1}_E|(W) (= \operatorname{vol}^n(\partial E \cap W) \text{ when } \partial E \text{ is smooth}).$$

So, we can define:

Definition 4.2. Let $E \subseteq \mathbb{R}^{n+1}$ be open. We say that E is a *Caccioppoli set* if it has locally finite perimeter: that is, $\mathbb{1}_E \in \text{BV}_{\text{loc}}$.

A Caccioppoli set E is *minimizing* if for every compact $W \subseteq \mathbb{R}^{n+1}$ and Caccioppoli set F agreeing with E outside of W (i.e., $E \Delta F \subseteq W$),

$$P(E, W) = |D\mathbb{1}_E|(W) \leq |D\mathbb{1}_F|(W) = P(F, W).$$

4.2. Existence. Our second task is to prove that within the class of Caccioppoli sets, we can make sense of limits. Recall that we had Γ an entire minimal graph, and were looking at the sequence $\lambda\Gamma$ as $\lambda \rightarrow \infty$, which is a sequence of minimizing Caccioppoli sets. The following compactness theorem ensures that limits exist:

Theorem 4.3 (Compactness for Caccioppoli sets). *Let $E_j \subseteq \mathbb{R}^{n+1}$ be a sequence of Caccioppoli sets with uniform perimeter bounds on compact sets: for all $W \subseteq \mathbb{R}^{n+1}$ compact, there exists $C(W) > 0$ such that*

$$P(E_j, W) \leq C(W)$$

for all j . Then there exists a Caccioppoli set E_∞ such that up to a subsequence, $E_j \rightharpoonup^ E_\infty$ in the following sense:*

$$\mathbb{1}_{E_j} \rightarrow \mathbb{1}_{E_\infty} \text{ in } L^1_{\text{loc}} \quad \text{and} \quad |D\mathbb{1}_{E_j}| \rightharpoonup^* |D\mathbb{1}_{E_\infty}| \text{ as Radon measures.}$$

This result follows immediately from compactness of BV functions, which is the exact same result but with $\mathbb{1}_{E_j}$ replaced by an arbitrary sequence of BV_{loc} functions u_j , and with uniform perimeter bounds on compact sets replaced with uniform L^1_{loc} and variation bounds on compact sets. In our setting, uniform L^1_{loc} bounds come for free, since $\|\mathbb{1}_E\|_{L^1(K)} \leq |K|$.

The analogous result of compactness for BV functions follows from a standard argument:

- (1) Mollification by radial bump functions gives smooth approximation of $u \in BV_{\text{loc}}$ by $u^{(\sigma)} \in C^\infty$ with $u^{(\sigma)} \rightarrow u$ in L^1_{loc} . By semicontinuity of the variation, we also have $|Du^{(\sigma)}| \rightharpoonup^* |Du|$ as Radon measures.
- (2) Rellich compactness for $W^{1,1}_{\text{loc}} \Subset L^1_{\text{loc}}$ gives compactness for the smooth approximations.
- (3) A diagonal argument gives compactness for BV_{loc} .

In our setting of blowdowns of minimal graphs, perimeter bounds are equivalent to area growth bounds, and hence we get compactness for free. Therefore, there exists Γ_∞ such that for some sequence $\lambda_j \nearrow \infty$, $\lambda_j\Gamma \rightharpoonup^* \Gamma_\infty$. Note that *a priori*, this limit is not unique: this is the cause of many headaches in the field.

4.3. Regularity. The final task is to prove regularity theorems. In the minimizing Caccioppoli set setting, we actually have fairly good regularity:

Theorem 4.4 (De Giorgi's regularity theorem). *Let $E \subseteq \mathbb{R}^{n+1}$ be a minimizing Caccioppoli set. Then, ∂E splits as a disjoint union*

$$\partial E = \text{reg } E \cup \text{sing } E,$$

where $\text{reg } E$ is a countable union of embedded C^1 hypersurfaces and $\text{sing } E$ has $\mathcal{H}^n$ measure zero.

For a minimizing Caccioppoli set E and $x \in E$, we can also consider the set of *tangent cones* at x :

$$C_x E = \{\partial F : \exists \lambda_j \searrow 0 \text{ such that } \lambda_j(E - x) \rightharpoonup^* F\}.$$

If $x \in \text{reg } E$, then $C_x E = \{T_x(\partial E)\}$ is the classical tangent space; if $x \in \text{sing } E$, then for a similar reason as above, $C_x E$ may not be unique.

4.4. Monotonicity. Recall that we want to show that if Γ is a nonflat minimal graph, Γ_∞ is a nonflat minimizing cone. The fact that Γ_∞ is minimizing follows from the fact that it is a limit of minimizers, but the other two claims require the following.

Theorem 4.5 (Monotonicity formula). *Denote by $\Theta_{x,r}(\Gamma)$ the density of a minimal hypersurface $\Gamma^n \subseteq \mathbb{R}^{n+1}$:*

$$\Theta_{x,r}(\Gamma) = \frac{\text{vol}^n(\Gamma \cap B(x, r))}{\omega_n r^n},$$

where $\omega_n = \text{vol}^n(B^n(0, 1))$. The following statements are true:

- (1) For every $x \in \Gamma$, the map $r \mapsto \Theta_{x,r}(\Gamma)$ is monotone nondecreasing.
- (2) $\Theta_{x,r}$ is constant in r if and only if Γ is a cone about x .
- (3) If Γ is a cone about x , then $\Theta_x = 1$ if and only if Γ is a plane.

From this, we get that Γ_∞ is a cone: at $x = 0$,

$$\Theta_r(\Gamma_\infty) = \lim_{\lambda \rightarrow \infty} \Theta_r(\lambda\Gamma) = \lim_{\lambda \rightarrow \infty} \Theta_{\lambda r}(\Gamma) = \Theta_\infty(\Gamma).$$

Moreover, this tells us that if Γ_∞ is flat, then Γ has constant density 1, and hence is a plane. Therefore, the contrapositive gives us that if Γ is nonflat, then so is Γ_∞ .

Note: there is a quantitative converse to the final statement: Allard's ε -regularity theorem says that there exists a constant $\varepsilon(n) > 0$ such that if C is a cone with density $\Theta < 1 + \varepsilon(n)$, then C is a plane.

5. FEDERER'S REDUCTION TO SMOOTH CONES AND SIMONS' CONE THEOREM

Now, we can see why Bernstein's theorem essentially comes down to classifying minimizing cones. But, there is a significant drawback to relying on minimizing cones *without smoothness assumptions*, since our techniques have to derive from geometric measure theory. If we want to utilize standard techniques in differential geometry, then we have to somehow work over a smooth manifold.

This is precisely what Federer (1970) accomplished. Note that if C is cylindrical, then $\text{sing } C = \text{sing } C' \times \mathbb{R}^l$, so C' has a smaller singular set than C . If C' is smooth, then it follows that $\text{sing } C = \{0\} \times \mathbb{R}^l$ is a linear subspace. This may not necessarily be so, but Federer shows that you can get there with a sequence of blowups.

Theorem 5.1 (Federer (1970)). *Let $C^n \subseteq \mathbb{R}^{n+1}$ be a non-flat non-smooth minimizing hypercone with $x \in \text{sing } C \setminus \{0\}$. Then, the blowup of C at x splits a line in the direction of x .*

Corollary 5.2. *Suppose $C^n \subseteq \mathbb{R}^{n+1}$ is a non-flat non-smooth minimizing hypercone. Then, there exists a non-flat smooth minimizing hypercone $\hat{C}^k \subseteq \mathbb{R}^{k+1}$ for $k < n$.*

This reduction to the smooth cone case, along with the following theorem of Simons (1968), completely resolves Bernstein's problem for $n \leq 7$.

Theorem 5.3 (Simons (1968)). *Let $n \in \{2, \dots, 6\}$. Let $\Sigma^{n-1} \subseteq S^n$ be a smooth minimal hypersurface with cone $C = C(\Sigma) \subseteq \mathbb{R}^{n+1}$. If C is stable, then C is flat.*

This completely resolves Bernstein's problem for $n \leq 7$:

- (1) Let $u: \mathbb{R}^n \rightarrow \mathbb{R}$ be a non-affine solution to the MSE, such that $\Gamma = \Gamma_u$ is a non-flat minimal graph.
- (2) Let $C^n \subseteq \mathbb{R}^{n+1}$ be a blowdown cone of Γ . By Observation 1, C is minimizing and nonflat.
- (3) By Federer's cone reduction theorem, up to taking a finite sequence of blowups, there exists a *smooth* minimizing hypercone $\widehat{C}^k \subseteq \mathbb{R}^{k+1}$ for $k < n \leq 7$; i.e., $k \leq 6$.
- (4) But, $\widehat{C}$ is minimizing and hence stable, so Simons's theorem says that $\widehat{C}$ is flat, a contradiction.

In fact, refining Federer and Simons's theorems, we get that a minimizing hypercone has singular set of Hausdorff codimension at least 7.

This is sharp: the Simons cone with $p = 4$

$$C_4 = \{(x, y) \in \mathbb{R}^4 \times \mathbb{R}^4 : |x|^2 = |y|^2\}$$

is minimizing. Indeed, Bombieri–de Giorgi–Giusti (1969) found a function $u: \mathbb{R}^8 \rightarrow \mathbb{R}$ such that Γ_u is nonflat, and its blowdown is $C_3 \times \mathbb{R}$.

6. SIMONS' CONE THEOREM

I will spend the last portion of my talk explaining Simons' cone theorem.

Recall that stability of a hypersurface $\Gamma^n \subseteq \mathbb{R}^{n+1}$ can be phrased quantitatively in terms of the nonnegativity of the first eigenvalue of the stability operator

$$L_\Gamma = \Delta_\Gamma + |A_\Gamma|^2,$$

which is in some sense the Hessian of the area functional.

Consider $\Sigma^{n-1} \subseteq S^n$ as in the theorem statement; the first eigenvalue of the stability operator for Σ is

$$\lambda_1(L_\Sigma) = \inf_{\varphi \in C^\infty(\Sigma) \setminus \{0\}} \frac{-\langle \varphi, L_\Sigma \varphi \rangle}{\langle \varphi, \varphi \rangle} = \inf_{\varphi \in C^\infty(\Sigma) \setminus \{0\}} \frac{\int_\Sigma |\nabla \varphi|^2 - |A_\Sigma|^2 \varphi^2}{\int_\Sigma \varphi^2}.$$

Simons' cone theorem comes down to proving two inequalities for $\lambda_1(L_\Sigma)$.

Proposition 6.1. *Assuming that $C(\Sigma)$ is not flat, the first eigenvalue of the stability operator of the link satisfies the following lower and upper bounds.*

$$-\frac{(n-2)^2}{4} \leq \lambda_1(L_\Sigma) \leq -(n-1).$$

If $C(\Sigma)$ is flat, then $\lambda_1(L_\Sigma) = 0$.

Simons' theorem is a very quick corollary: assuming that $C(\Sigma)$ is not flat, these inequalities can only be jointly satisfied when

$$-\frac{(n-2)^2}{4} \leq -(n-1).$$

Rearranging, we see that this is equivalent to

$$0 \leq \frac{(n-2)^2}{4} - (n-1) = \frac{n^2 - 4n + 4}{4} - n + 1 = \frac{n^2}{4} - 2n + 2.$$

The quadratic formula yields that the polynomial on the right has roots

$$\frac{2 \pm \sqrt{4-2}}{\frac{1}{2}} = 4 \pm \sqrt{8},$$

where $4 - \sqrt{8} \approx 1.17$ and $4 + \sqrt{8} \approx 6.83$. Hence, if $2 \leq n \leq 6$, this is a contradiction.

We will first explain the lower bound. Let $L_C = \Delta_C + |A_C|^2$ denote the stability operator of the cone. For $w \in C^\infty(C)$ with $w > 0$, define the *weighted first eigenvalue* by

$$\lambda_1(L_C, w) = \inf_{\varphi \in C^\infty(C) \setminus \{0\}} \frac{-\langle \varphi, L_C \varphi \rangle}{\langle \varphi, w \varphi \rangle} = \inf_{\varphi \in C^\infty(C) \setminus \{0\}} \frac{-\int_C \varphi L_C \varphi}{\int_C w |\varphi|^2}.$$

Theorem 6.2. *Let $\Sigma^{n-1} \subseteq S^n$ be a smooth minimal hypersurface and let $C = C(\Sigma)$ be the cone over Σ . Then*

$$\lambda_1(L_C, r^{-2}) = \lambda_1(L_\Sigma) + \frac{(n-2)^2}{4}.$$

Sketch of the proof. The proof is a combination of a conformal change of coordinates and a separation of variables argument.

First perform a conformal change of coordinates on the cone: take spherical coordinates $(\theta, r) \in C$ and turn them into cylindrical coordinates $(\theta, t) \in \Sigma \times \mathbb{R}$ by taking $t = \log r$. Up to a conformal change in the metric, the resulting metric on the cylinder is diagonal: $r^{-2}g_C = g_\Sigma + dt^2$.

The cone has no radial curvature, so the second fundamental form in cylindrical coordinates is equal to the second fundamental form on the link, up to the conformal rescaling: $r^{-2}|A_C|^2 = |A_\Sigma|^2$. Note that the volume form in spherical coordinates is $r^{n-1} d\theta dr$, and since $r = e^t$, $dr = r dt$, so the volume form in cylindrical coordinates is $e^{nt} d\theta dt$. Combined with the conformal rescaling, the new volume form in cylindrical coordinates is

$$d\text{vol}_{\Sigma \times \mathbb{R}} = e^{(n-2)t} d\theta dt.$$

In these coordinates, given an ansatz of the form $\varphi(\theta, t) = \eta(\theta)\rho(t)e^{-\frac{n-2}{2}t}$ (to cancel out the factor of $e^{(n-2)t}$), the eigenvalue problem $L_C\psi + \lambda\psi = 0$ splits into the two eigenvalue problems on Σ and $\mathbb{R}$, respectively:

$$L_\Sigma \eta + \mu_1 \eta = 0 \quad \text{and} \quad \left(\partial_t^2 + (n-2)\partial_t - \frac{(n-2)^2}{4} \right) \rho + \mu_2 \rho = 0,$$

with $\lambda = \mu_1 + \mu_2$.

The first eigenvalue for the first problem is clearly $\mu_1 = \lambda_1(L_\Sigma)$. For the second problem, we can analyze each piece separately.

The first eigenvalue of ∂_t^2 is 0: consider a sequence of bump functions $\rho_R \in C_c^\infty(\mathbb{R})$ with $\rho \equiv 1$ on $[-R, R]$, $\rho \equiv 0$ outside of $[-2R, 2R]$, and $|\rho'| \leq 2R^{-1}$. Then

$$\int_{\mathbb{R}} \rho^2 \geq 4R \quad \text{and} \quad \int_{\mathbb{R}} |\rho'|^2 \leq \frac{2}{R} 4R = 8,$$

so

$$\frac{\int_{\mathbb{R}} |\rho'_R|^2}{\int_{\mathbb{R}} \rho_R^2} \leq \frac{8}{4R} \rightarrow 0$$

as $R \rightarrow \infty$.

The first eigenvalue of $(n-2)\partial_t$ is 0: this follows immediately from noticing that

$$\int_{\mathbb{R}} \rho \rho' = \frac{1}{2} \int_{\mathbb{R}} (\rho^2)' = 0$$

since ρ is compactly supported.

Finally, the first eigenvalue of $-\frac{(n-2)^2}{4}$ is $\frac{(n-2)^2}{4}$.

Therefore, we get that

$$\lambda_1(L_C, r^{-2}) = \lambda_1(L_\Sigma) + \frac{(n-2)^2}{4}.$$

□

The lower bound now follows from stability of C .

Now we move on to the upper bound. The idea is to plug in $\varphi = |A_\Sigma|$ into the stability inequality: recall that $\lambda_1(L_\Sigma)$ is the infimum over all λ such that

$$\int_{\Sigma} |\nabla_\Sigma \varphi|^2 - |A_\Sigma|^2 \varphi^2 \leq \lambda \int_{\Sigma} \varphi^2$$

holds for all $\varphi \in C^\infty(\Sigma)$. Hence, if we can prove that this inequality is achieved for some λ , we get an upper bound $\lambda_1(L_\Sigma) \leq \lambda$.

To compute the LHS of the above expression, we need to do some calculations. The key idea is contained in the following elementary derivation of *Bochner's identity*. Let $u \in C^3(\mathbb{R}^n)$.

$$\begin{aligned} \frac{1}{2} \Delta |\nabla u|^2 &= \frac{1}{2} \sum_{i,j} \partial_i^2 (u_j^2) \\ &= \sum_{i,j} \partial_i (u_j u_{ji}) \\ &= \sum_{i,j} u_{ji}^2 + u_j u_{jii} \\ &= \sum_{i,j} u_{ji}^2 + u_j u_{iij} + u_j (u_{jii} - u_{iij}) \\ &= |\nabla^2 u|^2 + \langle \nabla u, \nabla \Delta u \rangle + \langle \nabla u, \text{curvature terms} \rangle. \end{aligned}$$

But now note that

$$\frac{1}{2} \Delta w^2 = \frac{1}{2} \sum_i \partial_i^2 (w^2) = \sum_i \partial_i (w w_i) = \sum_i (w_i)^2 + w w_{ii} = |\nabla w|^2 + w \Delta w,$$

so substituting $w = |\nabla u|$ we get

$$|\nabla u| \Delta |\nabla u| = \left(|\nabla^2 u|^2 - |\nabla |\nabla u||^2 \right) + \langle \nabla u, \nabla \Delta u \rangle + \langle \nabla u, \text{curvature} \rangle$$

Now substitute $V = \nabla u$:

$$|V| \Delta |V| = (|\nabla V|^2 - |\nabla |V||^2) + \langle V, \nabla \operatorname{div} V \rangle + \langle V, \text{curvature} \rangle.$$

Practically the same calculation, with a lot more indices, gives an analogous identity for $|A_\Sigma| = |\nabla_\Sigma \nu|$:

Proposition 6.3 (Simons' identity). *Let $\Sigma^{n-1} \subseteq S^n$ be a minimal hypersurface. Then,*

$$|A_\Sigma| \Delta_\Sigma |A_\Sigma| = (|\nabla A_\Sigma|^2 - |\nabla |A_\Sigma||^2) - |A_\Sigma|^4 + (n-1) |A_\Sigma|^2.$$

This then gives us that

$$|A_\Sigma| L_\Sigma |A_\Sigma| = |A_\Sigma| \Delta_\Sigma |A_\Sigma| + |A_\Sigma|^4 = (|\nabla A_\Sigma|^2 - |\nabla |A_\Sigma||^2) + (n-1) |A_\Sigma|^2.$$

We can prove that the term in parentheses is nonnegative:

$$|\nabla A_\Sigma|^2 - |\nabla |A_\Sigma||^2 \geq 0,$$

and hence we are left with the inequality

$$|A_\Sigma| L_\Sigma |A_\Sigma| \geq (n-1) |A_\Sigma|^2.$$

Integrating both sides, we get that

$$-\int_\Sigma |A_\Sigma| L_\Sigma |A_\Sigma| \leq -(n-1) \int_\Sigma |A_\Sigma|^2.$$

assume that $C(\Sigma)$ is not flat, so that $\int_\Sigma |A_\Sigma|^2 > 0$. Then, dividing by the L^2 -mass of the second fundamental form, we see that

$$\lambda_1(L_\Sigma) \leq -(n-1).$$

APPENDIX A. SCHRÖDINGER OPERATORS ON BOUNDED DOMAINS IN $\mathbb{R}^n$

In another version of this talk, I would have spoken in greater detail about the variational aspect of the minimal surface equation, by way of comparison with Schrödinger eigenfunctions, which are slight generalizations of Laplace eigenfunctions.

Let $\Omega \subseteq \mathbb{R}^n$ be a *smooth domain* (connected open subset with $\partial\Omega$ smooth) and let $V \in C^\infty(\Omega)$ be a smooth potential. The *Schrödinger operator* associated to Ω and V is $L = \Delta + V$, and the associated *eigenvalue problem with vanishing Dirichlet boundary conditions* in the *strong formulation* is

$$(*) \quad \begin{cases} Lu + \lambda u = 0 & \text{on } \Omega \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

for $u \in C^2(\Omega) \cap C(\overline{\Omega})$ and $\lambda \in \mathbb{R}$ unknown.

Now let us consider the *weak formulation* of this PDE: for a test function $\varphi \in C_0^\infty(\Omega)$, integrating both sides of the PDE against φ gives

$$0 = \int_{\Omega} \varphi (\Delta u + Vu + \lambda u) = \int_{\Omega} -\langle \nabla \varphi, \nabla u \rangle + \varphi Vu + \varphi \lambda u.$$

Therefore, we can make sense of the PDE for any $u \in W_0^{1,2}(\Omega)$, where

$$W_0^{1,2}(\Omega) = \left\{ u : \|u\|_{W^{1,2}} := \|u\|_{L^2(\Omega)} + \|Du\|_{L^2(\Omega)} < \infty \text{ and } u|_{\partial\Omega} = 0 \right\}.$$

By standard elliptic regularity theory, we can show that

Theorem A.1 (Regularity of Schrödinger eigenfunctions). *Let $\Omega \subseteq \mathbb{R}^n$ be a smooth domain and $V \in C^\infty(\Omega)$. Then, there exist $\lambda_k \in \mathbb{R}$ (with multiplicity) and $u_k \in C^\infty(\Omega) \cap W_0^{1,2}(\Omega)$ such that*

$$Lu_k + \lambda_k u_k = 0.$$

In fact, the same is true over any (Ω, g) compact Riemannian manifold with boundary.

Let us now give a second characterization of the PDE (*).

Definition A.2 (Dirichlet energy). The *Dirichlet energy* $\mathcal{E} : W_0^{1,2}(\Omega) \rightarrow \mathbb{R}$ is

$$\mathcal{E}(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 - Vu^2.$$

We would like to calculate the “gradient” of $\mathcal{E}$, but of course $W_0^{1,2}$ is infinite-dimensional. Recall that if $f : \mathbb{R}^n \rightarrow \mathbb{R}$, then

$$D_v f(x) := \left. \frac{d}{dt} \right|_{t=0} f(x + tv) = \langle v, \nabla f(x) \rangle.$$

Instead of trying to take the gradient of $\mathcal{E}$ all at once, we first compute the time-derivative on paths in the direction of a given function—this is called a *variation*. To this end, let us consider the path $u + t\varphi \in W_0^{1,2}(\Omega)$, for $\varphi \in C_c^\infty(\Omega)$ (consider this a dense subspace of the

full tangent space $W_0^{1,2}(\Omega)$). Computing the time-derivative, we get the *first variation in the direction of φ*

$$\begin{aligned}
\delta_\varphi \mathcal{E}(u) &:= \left. \frac{d}{dt} \right|_{t=0} \frac{1}{2} \int_\Omega |\nabla(u + t\varphi)|^2 - V(u + t\varphi)^2 \\
&= \frac{1}{2} \int_\Omega \left. \frac{d}{dt} \right|_{t=0} \langle \nabla u + t\nabla\varphi, \nabla u + t\nabla\varphi \rangle - V(u^2 + 2tu\varphi + t^2\varphi^2) \\
&= \int_\Omega \langle \nabla\varphi, \nabla u + t\nabla\varphi \rangle - V(u\varphi + t\varphi^2) \Big|_{t=0} \\
&= \int_\Omega \langle \nabla\varphi, \nabla u \rangle - Vu\varphi \\
&= - \int_\Omega \varphi (\Delta u + Vu) \\
&= - \langle \varphi, Lu \rangle_{L^2}
\end{aligned}$$

This is the *first variation formula*:

Proposition A.3 (First variation formula). *Let $u \in W_0^{1,2}(\Omega)$ and $\varphi \in C_c^\infty(\Omega)$. Then, the first variation of $\mathcal{E}$ in the direction of φ is*

$$\delta_\varphi \mathcal{E}(u) = - \langle \varphi, Lu \rangle_{L^2}.$$

Therefore, the gradient of $\mathcal{E}$ at u is $-L$.

We say that a function u is *stationary for $\mathcal{E}$* or a *critical point of $\mathcal{E}$* if $\delta_\varphi \mathcal{E}(u) = 0$ for all $\varphi \in C_c^\infty(\Omega)$.

It follows that

$$u \text{ minimizes } \mathcal{E} \text{ (i.e., } \mathcal{E}(u) = \inf_{W_0^{1,2}(\Omega)} \mathcal{E}) \implies u \text{ is stationary for } \mathcal{E} \text{ (i.e., } Lu = 0).$$

Of course, we should not expect the converse to be true, since critical points need not be extrema. Indeed, to refine this implication, we introduce the notion of a *second variation*, which is akin to the Hessian of $\mathcal{E}$.

Recall that if $f: \mathbb{R}^n \rightarrow \mathbb{R}$, then

$$D_v^2 f(x) := \left. \frac{d^2}{dt^2} \right|_{t=0} f(x + tv) = \left. \frac{d}{dt} \langle v, \nabla f(x + tv) \rangle \right|_{t=0} = \langle v, \nabla^2 f(x + tv)v \rangle \Big|_{t=0} = \langle v, \nabla^2 f(x)v \rangle.$$

That is, the second derivative along a path of velocity v is the Hessian (considered as a quadratic form) evaluated on that vector.

Computing the second variation of $\mathcal{E}$,

$$\begin{aligned}
\delta_\varphi^2 \mathcal{E}(u) &:= \left. \frac{d^2}{dt^2} \right|_{t=0} \mathcal{E}(u + t\varphi) \\
&= \left. \frac{d}{dt} \int_\Omega \langle \nabla\varphi, \nabla u + t\nabla\varphi \rangle - V(u\varphi + t\varphi^2) \right|_{t=0}
\end{aligned}$$

$$\begin{aligned}
&= \int_{\Omega} \langle \nabla \varphi, \nabla \varphi \rangle - V \varphi^2 \Big|_{t=0} \\
&= - \int_{\Omega} \varphi \Delta \varphi + V \varphi^2 \\
&= - \langle \varphi, L \varphi \rangle_{L^2}.
\end{aligned}$$

This is the *second variation formula*:

Proposition A.4 (Second variation formula). *Let $u \in W_0^{1,2}(\Omega)$ and $\varphi \in C_c^\infty(\Omega)$. Then, the second variation of $\mathcal{E}$ in the direction of φ is*

$$\delta_\varphi^2 \mathcal{E}(u) = - \langle \varphi, L \varphi \rangle_{L^2}.$$

Therefore, the Hessian of $\mathcal{E}$ at u is (also) $-L$.

Now, recall that $x \in \mathbb{R}^n$ is a local minimizer of $f: \mathbb{R}^n \rightarrow \mathbb{R}$ if $D^2 f(x)$ is positive semidefinite: that is,

$$D_v^2 f(x) = \langle v, \nabla^2 f(x) v \rangle \geq 0$$

for all $v \in \mathbb{R}^n$. We can phrase this quantitatively using the following formula for the first eigenvalue of a symmetric matrix A : if $\lambda_1 \leq \dots \leq \lambda_n$ are the eigenvalues of A , then

$$\lambda_1(A) = \inf_{v \neq 0} \frac{\langle v, Av \rangle}{\langle v, v \rangle}.$$

Hence, A is positive semidefinite if and only if $\lambda_1(A) \geq 0$.

We can do the same in the infinite dimensional case.

Definition A.5 (First eigenvalue). The *Rayleigh quotient for the first (Dirichlet) eigenvalue of L* is

$$\lambda_1(L) = \inf_{\varphi \in C_c^\infty(\Omega) \setminus \{0\}} \frac{\langle \varphi, -L \varphi \rangle_{L^2}}{\langle \varphi, \varphi \rangle_{L^2}}.$$

We say that u is *stable* (local minimum) if $\delta^2 \mathcal{E}(u)$ is positive semidefinite, which in this case is equivalent to $\lambda_1(L) \geq 0$. Note that in this case, since $\delta^2 \mathcal{E}(u) = -L$ does not depend on u , once we know that $\lambda_1(L)$ is nonnegative, we have that $\mathcal{E}$ is convex. Hence, $\mathcal{E}$ -minimizers are equivalent to $\mathcal{E}$ -stationary functions.